

Arunava Roy

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Overview

Data Engineer with 3+ years of hands-on experience building and optimizing large-scale batch pipelines on AWS, with additional exposure to DevOps and IaC practices. Actively seeking new opportunities to contribute to high-impact data engineering teams.

TECHNICAL SKILLS

Languages/Query: Python, PySpark, SQL.

AWS Cloud: S3, Glue, Lambda, Redshift, Athena, Managed Airflow, EC2, SNS, SQS, EventBridge, IAM.

Data Warehousing: Amazon Redshift, Athena, Data Modeling, Partitioning, SCD Type 1/2.

Infra And DevOps: Terraform, CloudFormation (SAM), CI/CD, SonarQube, Linux.

Version Control: Git, GitHub, GitLab.

Experience

Software Engineer, Zensar Technologies, Project Role: Data Engineer: Sept 2022 – Present

- Contributed to a multi-stage batch data pipeline spanning an AWS-based data lake and a Redshift data warehouse, processing 1–1.5 GB of Parquet data across two scheduled daily runs — ingesting from multiple heterogeneous source systems (Infomix, Oracle, AWS) via Glue Crawlers and Apache Hudi, through to warehouse-layer transformations and secure delivery to the end client via a privately shared S3 bucket, serving hundreds of retail store locations.
- Owned end-to-end DDL and DML development across the Raw ingestion layer (Athena) and four Redshift warehouse layers covering structured ingestion, delta detection via MD5 hashing, client-specific business transformations, and historical archival across 1–2 interfaces and approximately 15 tables.
- Reduced batch processing time by 50 percent (from 1 hour to 30–35 minutes) through targeted SQL query optimization, schema tuning, and Glue job configuration improvements.
- Designed and productionized an SNS-based source freshness alerting system, originally a POC which triggers notifications if no file arrives in a designated S3 path within 24 hours, preventing silent pipeline failures.
- Maintained daily data quality count checks across all pipeline stages, publishing results to a monitoring dashboard to track load consistency and detect volume anomalies for downstream teams.

DevOps & Platform Engineering:

- Authored Terraform scripts to provision and manage AWS infrastructure including EC2 instances, VPCs, Elastic IPs, and Lambda functions across Dev, QA, and Prod environments in a monorepo GitHub setup, ensuring consistent and repeatable deployments.
- Integrated SonarQube as a CI quality gate into the GitHub Actions workflow, enabling automated early detection of code smells and technical debt, improving team-wide code quality standards.
- Managed multi-environment CI/CD pipelines supporting the platform team's release cycles, improving deployment reliability and reducing manual intervention.

Education

Techno India University, BTech in Electronics And Communication Engineering Aug 2018 – Jun 2022

- CGPA: 8.84/10

Courses & Certifications

AWS Certified AI Practitioner – AWS:

- Gained foundational knowledge of AI/ML and generative AI concepts, with hands-on exposure to AWS AI services including SageMaker, Rekognition, Comprehend, and Bedrock etc.